LAB 1

meta pitt neurom (and&or)

[29]

```python
import numpy as np

def mc_pitts_neuron(x, w, theta):
    return 1 if np.dot(x, w) >= theta else 0
print("-----And Gate-----")
# AND gate parameters
w_and = np.array([1, 1])
theta_and = 2

for x in [(0, 0), (0, 1), (1, 0), (1, 1)]:
    print(x, mc_pitts_neuron(np.array(x), w_and, theta_and))
print("-----OR Gate-----")
# OR gate parameters
w_or = np.array([1, 1])
theta_or = 1

for x in [(0, 0), (0, 1), (1, 0), (1, 1)]:
    print(x, mc_pitts_neuron(np.array(x), w_or, theta_or))
```

```python
for x in [(0, 0), (0, 1), (1, 0), (1, 1)]:
    print(x, mc_pitts_neuron(np.array(x), w_or, theta_or))


print("Azeez zaidi 1/23/set/bcs/418")
```

```
-----And Gate-----
(0, 0) 0
(0, 1) 0
(1, 0) 0
(1, 1) 1
-----OR Gate-----
(0, 0) 0
(0, 1) 1
(1, 0) 1
(1, 1) 1
Azeez zaidi 1/23/set/bcs/418
```

FOR 3 INPUTS

```python
import numpy as np

def mc_pitts_neuron(x, w, theta):
    return 1 if np.dot(x, w) >= theta else 0
```

```python
import numpy as np
def art_neuron(x,w,b, activation='step'):
  z= np.dot(x,w)+b
  if activation=='step':
    return 1 if z>=0 else 0
  elif activation=='sigmoid':
    return 1/(1+np.exp(-z))
  elif activation == 'relu':
    return max(0,z)
  else:
    return"Invalid activation function"
x=np.array([1,2,3])
w=np.array([0.1,0.2,0.3])
b=0.1
print(art_neuron(x,w,b))

print("--------------------")
print("Step Output     :", art_neuron(x, w, b, activation='step'))
print("--------------------")
print("Sigmoid Output     :", art_neuron(x, w, b, activation='sigmoid'))
print("--------------------")
print("ReLu Output     :", art_neuron(x, w, b, activation='relu'))
print("--------------------")
```

FOR 3 INPUTS

```python
import numpy as np

def mc_pitts_neuron(x, w, theta):
    return 1 if np.dot(x, w) >= theta else 0
print("-----And Gate-----")
# AND gate parameters
w_and = np.array([1, 1, 1])
theta_and = 3

for x in [(0, 0, 0), (0, 0, 1), (0, 1, 0), (1, 0, 0),(1, 0, 1),(1, 1, 0),(1, 1, 1)]:
    print(x, mc_pitts_neuron(np.array(x), w_and, theta_and))
print("-----OR Gate-----")
# OR gate parameters
w_or = np.array([1, 1, 1])
theta_or = 1

for x in [(0, 0, 0), (0, 0, 1), (0, 1, 0), (1, 0, 0),(1, 0, 1),(1, 1, 0),(1, 1, 1)]:
    print(x, mc_pitts_neuron(np.array(x), w_or, theta_or))
```

```
-----And Gate-----
(0, 0, 0) 0
(0, 0, 1) 0
(0, 1, 0) 0
(1, 0, 0) 0
```

0s

```python
print(art_neuron(x,w,b))

print("--------------------")
print("Step Output      :", art_neuron(x, w, b, acti
print("--------------------")
print("Sigmoid Output      :", art_neuron(x, w, b, a
print("--------------------")
print("ReLu Output      :", art_neuron(x, w, b, acti
print("--------------------")
```

```
1
--------------------
Step Output      : 1
--------------------
Sigmoid Output      : 0.8175744761936437
--------------------
ReLu Output      : 1.5
--------------------
```